

Notes: Goldberg, Khandelwal, Pavcnik & Topalova (QJE, 2010)
Imported Intermediate Inputs and Domestic Product Growth: Evidence from India

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1 Big picture

- **Question.** Do cheaper and newly available imported intermediate inputs cause domestic firms to expand the range of products they manufacture?
- **Setting.** The paper studies India during the trade liberalization of the 1990s, a period with large and uneven reductions in input tariffs across industries.
- **Main empirical object.** The dependent variable is firm product scope: the number of products a domestic firm reports manufacturing in Prowess.
- **Main policy variation.** Input tariffs are constructed from India's input-output table: each final-good industry's input tariff is a weighted average of tariffs on the inputs used in that industry.
- **Main reduced-form result.** A 10 percentage point decline in input tariffs is associated with roughly a 3.2% increase in firm product scope. Given actual tariff declines, input tariff reductions explain about 31% of the expansion in firm product scope.
- **Mechanism.** The paper decomposes import-price changes into a conventional price channel and a variety channel. The variety channel is the main channel: new imported input varieties relax technological constraints faced by domestic producers.
- **Static gains.** Imported intermediate varieties lower the exact import price index, especially for intermediate inputs. The import price index for intermediate sectors is estimated to be about 4.75% lower per year due to new varieties.
- **Dynamic gains.** Access to new imported inputs is linked to new domestic products. This is a microeconomic channel through which trade liberalization can generate growth.
- **Economic takeaway.** Trade liberalization affects domestic production not only by making existing inputs cheaper, but also by expanding the input set available to firms. In India, this extensive margin of imported intermediates was quantitatively central.

2 Incremental contribution of the paper

- **Moves from imported varieties to domestic product creation.** Earlier work measured gains from imported varieties, but this paper directly links imported input availability to the creation of new domestic product lines by firms.
- **Focuses on input tariffs rather than output tariffs.** The paper shifts attention away from final-good competition and toward imported intermediates as a production technology shock.
- **Uses firm-level product-scope data.** The paper's Prowess data allow the authors to observe domestic product additions rather than using export varieties or industry R&D as indirect proxies.
- **Separates price and variety channels.** The Feenstra/Broda-Weinstein exact price-index methodology lets the authors distinguish lower prices of existing imported inputs from the arrival of new imported input varieties.
- **Builds sector-level input price indices.** By combining detailed Indian import data with the input-output table, the paper converts product-level import changes into industry-level input-cost shocks relevant for domestic firms.
- **Quantifies an endogenous-growth mechanism.** The estimates make concrete a channel emphasized in Romer-style and Grossman-Helpman-style models: imported intermediates increase the feasible set of domestic production activities.
- **Addresses reverse causality.** Domestic product growth could itself cause more input imports. The paper uses tariff changes and proximity-based instruments to separate tariff-induced input access from firm-driven import demand.

- **Offers a growth-accounting interpretation.** Since product additions accounted for about a quarter of Indian manufacturing output growth in the authors’ companion work, the estimated 31% contribution of input tariffs to product-scope growth implies a meaningful aggregate channel.
- **Bridges trade, firm scope, and industrial development.** The paper shows that import liberalization can raise domestic variety production in a developing economy, rather than only shifting market shares among existing firms and products.

3 Data and measurement

3.1 Firm-level Prowess data

- The firm-level data come from Prowess, collected by the Centre for Monitoring the Indian Economy.
- Prowess includes relatively large Indian manufacturing firms and contains product-level information.
- The paper focuses on firms observed during 1989–1997, the period around India’s major trade reform.
- The key firm-level outcome is product scope: the number of products manufactured by firm i in industry q at time t .
- Because product dropping is rare in the sample, increases in product scope can be interpreted mainly as product additions.

Interpretation: The firm-level product data are crucial because they let the authors measure domestic product creation directly rather than relying on aggregate output, productivity, or export proxies.

3.2 Import data and product classification

- Import data are official Indian import data from Tips Software Services.
- The data record approximately 10,000 manufactured products imported from around 160 countries between 1987 and 2000.
- Imports are observed at the eight-digit Harmonized System level.
- For descriptive decompositions, products are classified as intermediate goods or final goods using the Nouroz classification of India’s input-output matrix.

Interpretation: The import data are a census of products. This matters because if a variety has zero imports before reform, it was unavailable anywhere in India through formal import channels.

3.3 Input tariffs

The central policy variable is an input tariff for each output industry. It is constructed by applying the input-output matrix to product/industry tariffs:

$$\tau_{qt}^{input} = \sum_i \alpha_{iq} \tau_{it},$$

where α_{iq} is the cost share of input industry i in output industry q , and τ_{it} is the tariff on input i at time t .

Interpretation: This transforms product-level tariff reductions into industry-level shocks to the cost and availability of inputs used by domestic firms.

3.4 Output tariffs and other reforms

- The paper distinguishes input tariffs from output tariffs.
- Output tariffs measure protection on a firm's own final goods; input tariffs measure tariffs on the intermediates the firm uses.
- Robustness checks control for delicensing and FDI liberalization, because trade liberalization coincided with other Indian reforms.

Interpretation: The distinction matters because the paper's mechanism is not final-good competition. It is access to imported intermediates.

3.5 Exact import price and variety indices

The exact price index follows the Feenstra (1994) and Broda-Weinstein methodology. For a product group, the exact price index can be written schematically as

$$P_{gt}^{exact} = P_{gt}^{conv} \times V_{gt},$$

where P_{gt}^{conv} captures price changes of continuing varieties and V_{gt} captures the welfare effect of newly available varieties.

At the industry level, input-output weights aggregate input-group price indices into an industry input index:

$$P_{qt}^{input} = \prod_i (P_{it}^{exact})^{\alpha_{iq}}.$$

Interpretation: The decomposition lets the authors ask whether domestic product growth comes from cheaper continuing inputs or from access to inputs that did not previously exist in the Indian import bundle.

4 Models and empirical specifications

4.1 Baseline reduced-form product-scope regression

The main firm-level specification is

$$\ln(n_{iqt}) = \alpha_i + \alpha_t + \beta \tau_{qt}^{input} + \varepsilon_{iqt},$$

where n_{iqt} is the number of products made by firm i , α_i are firm fixed effects, and α_t are year fixed effects.

Interpretation: The coefficient β measures how firm product scope changes when the tariff on the firm's input bundle changes. A negative β means lower input tariffs increase product scope.

4.2 Import-value and import-variety regressions

The descriptive import regressions relate import outcomes to output tariffs:

$$\ln(M_{ht}) = \gamma_h + \gamma_t + \delta \tau_{h,t-1} + u_{ht},$$

where h denotes a product category. Similar regressions are run for unit values and for the number of varieties.

Interpretation: These regressions establish that tariff reductions were associated with more import value, lower import unit values, and more imported varieties, especially in intermediates.

4.3 Robustness and trend controls

The paper checks whether the baseline input-tariff effect survives:

- industry-year fixed effects at different levels of aggregation;
- interactions of pre-reform product/output/TFP/R&D growth with year fixed effects;
- Poisson specifications and long-difference specifications;
- controls for output tariffs, delicensing, and FDI liberalization.

Interpretation: These tests ask whether the effect is really an input-tariff effect rather than a spurious correlation with other reforms or pre-existing trends.

4.4 Structural channel regression

The mechanism section relates changes in firm product scope to the conventional price component and the variety component of the input price index:

$$\Delta \ln(n_i) = \theta_1 \Delta \ln P_q^{conv} + \theta_2 \Delta \ln V_q + u_i.$$

Interpretation: The comparison between θ_1 and θ_2 identifies whether product-scope expansion comes mostly from lower prices of existing inputs or from the new-input-variety channel.

4.5 Instrumental variables for channels

The IV specification uses input tariff changes and interactions with proximity-to-India measures as instruments for conventional input price and input variety components.

Interpretation: The IV strategy addresses the concern that sectors with growing product scope may endogenously attract new imported inputs. Tariffs and geographic/linguistic proximity shift input availability in ways plausibly unrelated to firm-level product-scope shocks.

5 Identification and empirical design

5.1 Core identification problem

The main endogeneity problem is reverse causality. Domestic firms may decide to introduce new products for non-trade reasons, and that product expansion may generate demand for new imported inputs. A simple correlation between imported input varieties and product scope could therefore overstate the causal effect of imported inputs.

5.2 Why Indian tariff reform helps

- India’s 1991 reforms reduced tariffs substantially and unevenly across products and industries.
- The initial post-1991 tariff changes were driven by balance-of-payments crisis and IMF-supported reform, not by lobbying by particular manufacturing firms.

- Pre-reform firm outcomes do not predict subsequent input tariff changes, as shown in Table I.
- The reform generated cross-industry variation in input tariff reductions after mapping product tariffs through the input-output table.

Interpretation: The tariff reform provides plausibly exogenous variation in access to imported intermediates across firms' input bundles.

5.3 Why input-output weights matter

The same tariff cut has different effects across domestic industries depending on which inputs they use. The input-output table converts import liberalization into industry-specific input-cost and input-variety shocks.

Interpretation: This is what makes the analysis about input access rather than general openness.

5.4 How the paper addresses alternative reforms

- Output tariffs are included to separate input access from final-good competition.
- Delicensing controls capture India's removal of industrial licensing restrictions.
- FDI liberalization controls capture changes in foreign investment policy.
- Robustness checks include industry-year interactions and pretrend controls.

Interpretation: The main input-tariff coefficient remains stable, suggesting that the product-scope results are not driven by other contemporaneous policy reforms.

5.5 Limitations of the design

- Firm-level data do not directly observe all imported inputs used by each firm.
- Firms may purchase imported intermediates through domestic intermediaries, making sector-level import measures preferable but less micro-direct.
- Tariffs vary at product/industry level, not at firm level, limiting within-industry causal variation.
- TFP interpretations are difficult in multiproduct firms because revenue productivity combines efficiency, markups, quality, and product mix.
- The paper identifies product-scope responses during India's 1990s reform; external validity depends on whether other countries had similarly restrictive import regimes before liberalization.

Interpretation: The evidence is strongest for the causal effect of input tariff reductions on product scope and for the importance of the variety channel; it is less direct on firm-level physical productivity or on aggregate welfare beyond the price-index calculations.

6 Section-by-section study guide

- **Introduction.** Establishes the question: imported inputs can generate both static gains through cheaper/more varied inputs and dynamic gains through new domestic products.
- **Data and policy background.** Describes Prowess, Indian import data, tariff data, and the construction of input tariffs using the input-output matrix.

- **Reduced-form results.** Shows that tariff reductions expanded imports and varieties, and that lower input tariffs increased firm product scope.
- **Mechanisms.** Builds exact input price indices and decomposes them into conventional price and variety components.
- **IV channel analysis.** Uses tariff and proximity instruments to show that imported input variety, rather than conventional price declines, is the main channel.
- **Conclusion.** Emphasizes that imported intermediates relax technological constraints and help domestic firms introduce new products.

7 Tables and results

7.1 Table I: Prereform Firm Characteristics and Input Tariff Changes + Table II: Decomposition of Import Growth, 1987–2000

Read:

- Table I tests whether 1997–1992 input tariff changes were correlated with pre-reform firm growth in products, output, TFP, or R&D.
- None of the prereform outcome coefficients is statistically significant, which supports the claim that tariff reductions were not targeted to pre-existing firm trends.
- Table II decomposes import growth into extensive and intensive margins.
- Total import growth is 130%; intermediate imports rise 227%, while final-goods imports rise 90%.
- For intermediate products, the net extensive margin contributes 153 percentage points, meaning new intermediate products account for roughly two-thirds of import growth.

Detailed interpretation: Table I is an identification table. It asks whether politicians selected industries for larger future tariff changes because those industries were already growing or declining. The lack of significant coefficients reduces this concern. Table II is a facts table. It shows that India’s trade liberalization was not just more of the same imported products; it radically expanded the set of intermediate products available to firms.

Economic message: The foundation of the paper is that input tariff changes were plausibly exogenous to prereform firm trends and that import growth was driven strongly by new intermediate varieties.

	Products (1)	Output (2)	TFP (3)	R&D (4)
1997–1992 input tariff change	0.180 (0.308)	−0.744 (0.745)	0.517 (0.507)	−1.531 (1.341)
Observations	713	712	614	667

Notes. The dependent variables in each column are the prereform 1989–1991 growth in firm-level outcomes. The variables are regressed on postreform (between 1992 and 1997) changes in input tariffs. Column (1) is the prereform firm-level change in (log) number of products. Columns (2)–(4) are the prereform changes in (log) firm output, TFP, and R&D expenditure. The number of observations varies in each column because the coverage of firm outcomes varies. Standard errors (in parentheses) clustered at the industry level.
* 10%, ** 5%, and *** 1% significance level.

(a) Table I: prereform characteristics and tariff changes

Product classification	Import growth (1)	Extensive margin			Intensive margin		
		Net (2)	Product entry (3)	Product exit (4)	Net (5)	Growing (6)	Shrinking (7)
All products	130	84	84	0	45	84	–39
Intermediate products	227	153	153	0	74	116	–42
Final products	90	33	33	0	57	86	–29

Notes. Real import growth decomposed into the extensive and intensive margins between 1987 and 2000. Imports are deflated by the wholesale price index. Column (1) reports overall import growth. Columns (2) and (5) report the contribution of the extensive and intensive margins, respectively. The extensive margin is growth in imports due to new six-digit HS codes not imported in 1987. The intensive margin measures import growth within products that India had imported in 1987. The gross contributions are reported in columns (3) and (4) for the extensive margin, and columns (7) and (8) for the intensive margin. Rows (2) and (3) decompose import growth in the intermediate (basic, capital, and intermediates) and final (consumer durables and nondurables) products. The HS codes have been standardized to remove any issues due to changes in the classification of HS codes.

(b) Table II: import growth decomposition

7.2 Table IIIa: Import Values and Tariffs + Table IIIb: Import Unit Values and Tariffs

Read:

- Table IIIa shows that lower output tariffs are associated with higher import values.
- The effect is significant for all products, intermediates, and final goods.
- Table IIIb shows that lower tariffs are associated with lower tariff-inclusive unit values.
- The coefficient is positive because a higher tariff raises unit values; therefore tariff cuts reduce import unit values.
- The intermediate coefficient in Table IIIb is especially large and statistically precise.

Detailed interpretation: Tables IIIa and IIIb document the two classic effects of tariff liberalization: more import value and lower import prices for continuing varieties. These are important controls for the paper’s main mechanism because they show that tariff reductions changed the trade environment before the authors isolate the variety channel.

Economic message: Tariff reductions increased imports and reduced prices, but these conventional channels do not exhaust the gains from liberalization.

	All products (1)	Intermediates (2)	Final goods (3)
Output tariff	−0.136*** (0.035)	−0.117*** (0.044)	−0.151** (0.076)
Year FEs	Yes	Yes	Yes
HS6 FEs	Yes	Yes	Yes
R ²	.82	.82	.80
Observations	35,833	20,140	11,838

Notes. Coefficients on tariffs from product-level regressions of log (fob) import value on lagged output tariffs, HS6 product fixed effects, and year effects. An observation is HS6-category-year. Column (1) pools across all countries. Columns (2) and (3) report coefficients for the intermediate and final goods, respectively.

(a) Table IIIa: import values and tariffs

	All products (1)	Intermediate (2)	Final goods (3)
Output tariff	0.273*** (0.050)	0.304*** (0.077)	0.245*** (0.079)
Year FEs	Yes	Yes	Yes
HS8-country FEs	Yes	Yes	Yes
R ²	.88	.85	.93
Observations	49,109	32,619	11,070

Notes. Regressions of (log) tariff-inclusive unit values on tariffs, HS8-country fixed effects, and year fixed effects. Unit values are computed for each HS8-country pair and the tariffs are the HS6 level. Column (1) uses all products and columns (2) and (3) report coefficients for the intermediates and final goods, respectively. Regressions are run from 1987 to 1997. Standard errors (in parentheses) clustered at the HS6 level.
*10%. **5%. and ***1% significance level.

(b) Table IIIb: import unit values and tariffs

7.3 Table IIIc: Import Extensive Margin and Tariffs + Table IVa: Product Scope and Input Tariffs

Read:

- Table IIIc shows that lower tariffs expand the number of imported varieties.
- The extensive-margin result is strongest for intermediate products and robust across several definitions of variety.
- Panel A defines variety as HS6-country; Panel B defines variety as HS8; Panel C defines variety as HS8-country.
- Table IVa shows the key firm-level reduced-form result: lower input tariffs increase firm product scope.
- The input tariff coefficient is about -0.32 in the baseline and remains negative and significant after controlling for output tariffs, delicensing, and FDI liberalization.

Detailed interpretation: Table IIIc provides the bridge from trade liberalization to new input availability. Table IVa then links the input tariff shock to firm behavior. The fact that output tariffs are insignificant while input tariffs matter is crucial: the paper is not primarily about final-goods competition, but about imported intermediates as productive inputs.

Economic message: The core reduced-form mechanism is already visible here: lower tariffs bring in new inputs, especially intermediates, and firms exposed to lower input tariffs expand product scope.

TABLE IIIc
IMPORT EXTENSIVE MARGIN AND TARIFFS

	All products (1)	Intermediate (2)	Final goods (3)
Panel A: Variety: HS6–country			
Output tariff	−0.082*** (0.012)	−0.106*** (0.014)	−0.049* (0.026)
R^2	.85	.84	.84
Observations	35,833	20,140	11,838
Panel B: Variety: HS8			
Output tariff	−0.015** (0.007)	−0.023*** (0.009)	−0.005 (0.014)
R^2	.88	.90	.85
Observations	35,833	20,140	11,838
Panel C: Variety HS8–country			
Output tariff	−0.095*** (0.013)	−0.129*** (0.016)	−0.042 (0.028)
R^2	.87	.86	.86
Observations	35,833	20,140	11,838

Notes. All regressions also include year fixed effects and HS6 fixed effects. The table reports coefficients on tariffs from product-level regressions of (log) number of varieties on output tariffs, HS6 product fixed effects, and year effects. The regressions are run at the HS6–year level and each panel uses an alternative definition of a variety. A variety is defined as an HS6–country pair in Panel A, an HS8 code in Panel B, and an HS8–country pair in Panel C. Within each panel, column (1) pools across all sectors, whereas columns (2) and (3) report coefficients for the intermediate and final goods, respectively. As in the previous tables, tariffs are at the HS6 level and the regressions are run from 1987 to 1997. Standard errors (in parentheses) clustered at the HS6 level.

*10%, **5%, and ***1% significance level.

(a) Table IIIc: import extensive margin and tariffs

TABLE IVa
PRODUCT SCOPE AND INPUT TARIFFS

	(1)	(2)	(3)	(4)
Input tariff	−0.323** (0.139)	−0.310** (0.150)	−0.327** (0.150)	−0.281** (0.125)
Output tariff		−0.013 (0.043)	−0.014 (0.041)	−0.010 (0.041)
Delicensed			−0.032 (0.023)	−0.026 (0.021)
FDI liberalized				0.037 (0.024)
Year effects	Yes	Yes	Yes	Yes
Firm FEs	Yes	Yes	Yes	Yes
R^2	.90	.90	.90	.90
Observations	14,882	14,864	13,435	11,135

Notes. The dependent variable in each regression is (log) number of products manufactured by the firm. The delicensed variable is an indicator variable obtained from Aghion et al. (2008) that switches to one in the year that the industry becomes delicensed. The FDI variable is a continuous variable obtained from Topalova and Khandelwal (2011), with higher values indicating a more liberal FDI policy. As with the tariffs, the licensed and FDI policy variables are lagged. All regressions include firm and year fixed effects and are run

(b) Table IVa: product scope and input tariffs

7.4 Table IVb: Product Scope and Input Tariffs - Reduced-Form Robustness Checks

Read:

- Table IVb subjects the baseline product-scope result to many robustness checks.
- The table controls for industry-year fixed effects at NIC2 and NIC3 levels.
- It interacts prereform industry product/output/TFP/R&D growth with year fixed effects.
- It includes specifications with initial products, initial firm output, initial output tariffs, initial TFP, initial R&D, Poisson models, and constant-firm samples.
- Long-difference checks yield coefficients close to the baseline.

Detailed interpretation: This is the paper’s main robustness table. The key idea is to progressively absorb alternative explanations: industry-specific shocks, pre-reform trends, initial firm characteristics, functional-form concerns, and sample-composition concerns. The persistence of the negative input-tariff coefficient supports a causal interpretation.

Economic message: The main finding is not fragile: the product-scope response survives a broad set of specifications designed to rule out obvious omitted-variable stories.

REDUCED-FORM ROBUSTNESS CHECKS			
Product scope regressed on input tariffs, firm and year fixed effects, plus controls			
(1) NIC2 $\times$ year FEs	-0.323*	(7) Initial products $\times$ year FEs	-0.408***
	(0.191)		(0.128)
(2) NIC3 $\times$ year FEs	-0.424**	(8) Initial output decile $\times$ year FEs	-0.311**
	(0.204)		(0.146)
(3) Prereform industry product growth $\times$ year FEs	-0.327**	(9) Initial TFP decile $\times$ year FEs	-0.321*
	(0.145)		(0.163)
(4) Prereform industry output growth $\times$ year FEs	-0.315**	(10) Initial R&D dummy $\times$ year FEs	-0.312**
	(0.141)		(0.154)
(5) Prereform industry TFP growth $\times$ year FEs	-0.336**	(11) Poisson model	-0.286*
	(0.141)		(0.162)
(6) Initial MPF dummy $\times$ year FEs	-0.393***	(12) Constant firms	-0.403*
	(0.127)		(0.206)
Long-difference robustness checks			
<i>Dependent variable: 1997–1989 change in log product scope</i>		<i>Dependent variable: 1997–1991 change in log scope—1991–1989 change in log scope</i>	
(13) Δ Input tariffs _{1997–1989}	-0.315	(14) Δ Input tariffs _{1997–1991}	-0.234
	(0.237)	Δ Input tariffs _{1991–1989}	0.251
Observations	696	Observations	662

Notes. The dependent variable in each regression is (log) number of products manufactured by the firm. Each row reports the coefficient on the input tariff with the additional controls beyond firm and year fixed effects. Specifications (1)–(12) have approximately 14,000 observations. Specifications (1) and (2) include two-digit NIC-year and three-digit NIC-year fixed effects, respectively. Specification (3) interacts the firms' initial multiple-product status with year fixed effects. Specification (4) interacts the firms' initial number of products with year fixed effects. Specifications (5) and (6) include interactions of year fixed effects with deciles of initial firm output and TFP. Specification (7) includes firms' initial R&D status with year dummies. Specifications (8)–(10) include interactions of year dummies with prereform (1989–1991) industry product, output and TFP growth, respectively. Specification (11) runs equation (3) using a Poisson regression using product scope (rather than log product scope) as the dependent variable. Specification (12) reruns equation (3) on a constant set of firms. The bottom panel reports long-difference specifications. Specification (13) runs a long-difference regression of the change in (log) scope on the change in input tariffs between 1989 and 1997. Specification (14) runs a long-difference regression that controls for firm-specific trends by running the change in log scope between 1991–1997 and 1989–1991 on the equivalent double difference in input tariffs. All regressions include firm and year fixed effects. Standard errors (in parentheses) clustered at the industry level. *10%, **5%, and ***1% significance level.

Table IVb: reduced-form robustness checks

7.5 Table IVc: Input Tariffs and Other Firm Outcomes + Table V: Import Variety Indices

Read:

- Table IVc shows that lower input tariffs are associated with higher output and higher TFP.
- The R&D coefficient is negative but imprecise overall; the interaction with large firms is negative and significant, implying a stronger R&D response among initially large firms.
- Table V reports import variety indices using Broda-Greenfield-Weinstein elasticities for India.
- The overall variety index is 0.688, implying large gains from variety relative to a conventional price index.
- Intermediate sectors have a lower overall variety index (0.624) than final sectors (0.850), meaning new intermediate varieties generate a larger price-index deflator.

Detailed interpretation: Table IVc expands the outcome space beyond product scope. It shows that the same policy shock is associated with firm growth and productivity changes, although the authors are cautious about interpreting revenue TFP. Table V quantifies the static gains from new varieties. The smaller the variety index, the larger the adjustment to the conventional price index from new varieties.

Economic message: The paper's story is not only that firms add products. Lower input tariffs are also linked to broader firm performance, and the variety gains are especially large for intermediate inputs.

TABLE IVc INPUT TARIFFS AND OTHER FIRM OUTCOMES				
	Output (1)	TFP (2)	R&D (3)	R&D (4)
Input tariff	-1.125** (0.436)	-0.454* (0.233)	-1.559 (1.751)	-0.077 (1.124)
Input tariff $\times$ large firm				-1.903* 1.111
Year effects	Yes	Yes	Yes	Yes
Firm FEs	Yes	Yes	Yes	Yes
R ²	.92	.81	.21	.21
Observations	14,874	13,714	14,233	14,233

(a) Table IVc: other firm outcomes

TABLE V IMPORT VARIETY INDICES			
	Variety index		
	Mean	Median	Overall
All sectors	0.899	0.986	0.688
Intermediate sectors	0.881	0.954	0.624
Final sectors	0.904	1.000	0.850

Notes. Table reports the variety index computed at the HS4 level using elasticities of substitution from Broda, Greenfield, and Weinstein (2006) for India. The indices use HS6-country pairs as the definition of a variety. Columns (1) and (2) report the median and mean variety index across HS4 groups. Column (3) aggregates the HS4 indices to the overall economy level using equation (13) in Broda and Weinstein (2006). The first row reports the variety index over all imported sectors. The second and third rows compute the indices for the intermediate and final sectors. The numbers are computed using data between 1989 and 1997.

(b) Table V: import variety indices

7.6 Table VIa: Product Scope and Channels - OLS + Table VIb: Product Scope and Channels - Instrumental Variables

Read:

- Table VIa relates firm scope to the conventional price index and the variety index.
- The variety index coefficient is large, negative, and statistically significant; the conventional price index coefficient is small and insignificant.
- Because the variety index falls when new varieties become more available, a negative coefficient means more input variety increases firm scope.
- Table VIb implements the IV strategy using input tariffs and proximity interactions.
- First-stage results confirm that tariffs and proximity shift conventional prices and variety access.
- Second-stage results again indicate that the variety channel is the dominant mechanism.

Detailed interpretation: Tables VIa and VIb are the paper's mechanism tables. They show that the reduced-form input-tariff effect operates through access to new imported varieties rather than merely cheaper continuing inputs. The IV table strengthens the interpretation by instrumenting the price-index components rather than treating them as exogenous.

Economic message: The mechanism is technological access. New imported inputs expand the feasible domestic product set, which is why liberalization affects product creation so strongly.

	(1)	(2)	(3)
Conventional price index	-0.156 (0.121)		-0.124 (0.113)
Variety index		-5.97** (2.55)	-5.70** (2.41)
R^2	.002	.009	.010
Observations	696	696	696

Notes: OLS regressions of firm scope on the imported input price indices. Columns (1) includes the conventional index, column (2) includes the variety index, and column (3) includes both indices. Regression is run for the years 1989 and 1997. Standard errors (in parentheses) clustered at the industry level.
*10%, **5%, and ***1% significance level.

(a) Table VIa: channels, OLS

	1st stage regressions		2nd stage regressions				
	{Conv. price index} _q (1)	{Variety index} _q (2)	{ΔFirm scope} _f (3)	{ΔFirm scope} _f (4)	{ΔFirm scope} _f (5)	{ΔFirm scope} _f —1998/99 IO matrix (6)	{ΔFirm scope} _f —polynomial instruments (7)
{ΔInput tariff} _{q,97-89}	1.340*** (0.476)	-0.007 (0.020)					
{ΔInput tariff} _{q,97-89} × {proximity} _q	-0.707 (1.618)	0.121* (0.065)					
{Conventional price index} _q			-0.240 (0.211)		-0.033 (0.299)	-0.047 (0.245)	-0.493* (0.271)
{Variety index} _q				-14.24* (7.56)	-13.40 (10.46)	-14.88* (8.97)	-14.80*** (5.00)
F -test 1st stage instruments			[5.3, $p = .01$]	[3.1, $p = .05$]	3.1, $p = .05$	5.5, $p = .01$	12.8, $p = .00$
Observations	696	696	696	696	696	696	696

Notes: Table reports IV regressions of firm scope on the imported input price indices. The instruments are changes in input tariffs and changes in input tariffs interacted with the proximity measure described in the text. Columns (1) and (2) report the first-stage regressions for the conventional price and variety indices, respectively. Columns (3)–(7) report the second-stage regressions. Column (6) uses the 1998/1999 input-output matrix. Column (7) includes third-order polynomials of the instruments and is estimated using a continuously updated OLS estimator. The regressions are run for the years 1989 and 1997. Standard errors (in parentheses) clustered at the industry level.
*10%, **5%, and ***1% significance level.

(b) Table VIb: channels, IV

8 Detailed result map

- **Prereform validation.** Future tariff changes are not predicted by pre-reform firm growth, helping validate the policy shock.
- **Import response.** Intermediate imports grow much more than final-good imports, and the extensive margin is much more important for intermediates.
- **Tariff-to-import mechanism.** Tariff cuts raise import values, reduce unit values, and expand imported varieties.
- **Firm-scope response.** Lower input tariffs significantly increase the number of products manufactured by Indian firms.

- **Robustness.** The reduced-form result survives controls for output tariffs, delicensing, FDI liberalization, pretrends, industry-year shocks, and sample restrictions.
- **Other outcomes.** Lower input tariffs are also associated with higher output, higher measured TFP, and larger-firm R&D responses.
- **Variety gains.** New imported varieties lower the exact import price index substantially, especially for intermediate sectors.
- **Mechanism.** The variety component of the input price index, not the conventional price component, is the key predictor of domestic product-scope growth.

9 Short summary

- The paper studies whether imported intermediate inputs spur domestic product growth.
- The setting is India’s 1990s trade liberalization.
- The main outcome is firm product scope in Prowess.
- The key policy shock is the input tariff, constructed using India’s input-output table.
- Pre-reform trends do not predict subsequent input tariff changes.
- Intermediate import growth is much larger than final-good import growth.
- New intermediate imported products account for a large share of import growth.
- Lower tariffs increase import values, lower import unit values, and expand imported varieties.
- Lower input tariffs increase firm product scope; the implied effect accounts for about 31% of firm product-scope growth.
- The effect is robust to many controls for other reforms and pre-existing trends.
- Exact price indices show large gains from new imported varieties, especially intermediate varieties.
- The channel analysis indicates that the variety channel, not the conventional price channel, drives the product-scope response.
- The paper provides micro evidence for a dynamic gains-from-trade mechanism emphasized by endogenous-growth theory.

10 Conclusion

10.1 Core conclusion

Goldberg, Khandelwal, Pavcnik, and Topalova show that access to imported intermediate inputs had a large effect on domestic product creation in India. Lower input tariffs increased the range of products produced by Indian firms, and the mechanism was primarily the arrival of new imported input varieties rather than only lower prices of existing imports.

10.2 Economic intuition

The intuition is that a firm cannot produce some goods without access to specialized inputs. When trade barriers are high, the domestic technology set is constrained by the limited local input bundle. Tariff liberalization expands the input set. Once firms can access new imported intermediates, they can introduce new domestic products that were previously infeasible or too costly.

10.3 Incremental contribution to trade and growth research

The paper makes the dynamic gains-from-trade mechanism measurable. It links a policy shock to input availability, input availability to firm product scope, and firm product scope to domestic product growth. This is stronger than showing that imported inputs raise productivity because product-scope changes are directly observed at the firm level.

10.4 Policy implication

Trade liberalization can foster domestic industrial growth when it reduces barriers on intermediate inputs. The result supports a policy distinction between input tariffs and output tariffs: protecting final goods may shield domestic producers, but taxing inputs can restrict their technological capabilities and inhibit product innovation.

10.5 What to remember

The paper's main lesson is that imports and domestic production are complements when imports are intermediates. Imported inputs can be a source of domestic product growth because they expand what domestic firms can make.

10.6 Limitations and open questions

The paper does not observe every input used by each firm, so the mechanism is measured through product-level import data and input-output weights. The welfare calculations focus on imported variety gains and product-scope responses, but do not provide a complete aggregate welfare accounting. Future work could use firm-input matched data to study which firms adopt which imported inputs and how those input choices affect product quality, markups, physical productivity, and long-run growth.